Path to profitability, a hospital financial analysis

A practice project by Luc Allen

The data in this project comes from a made up dataset, and there is no ties to real customer data.

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Summary

This report summarises an analysis of the revenue and performance of the hospital, with the aim to inform strategic growth initiatives and revenue optimisation. Data-driven insights into financial performance and optimization opportunities will be provided through analysis of key areas, including revenue generation by treatment type, price change modeling and others.

Through this analysis, it was discovered that the hospital has been operating at a loss, with the estimates putting it at around -£37,200 in the year 2023. This loss was driven primarily by failed payments, with only 32.9% of all bills being paid. To address this, two possible strategic initiatives have been suggested, and their effect simulated on the 2023 revenue. Implementing a reduced fee by 60% for failed payments, and introducing a 28% price increase to offset completed but unpaid appointments. Combining these two strategies could transform the 2023 loss of £37,200 into a profit of around £43,000. This will address the hospital's financial vulnerability and ensure its security going forward. The hospital was also analysed on a more operational level, in areas such as insurance provider reliability, doctor performance, and revenue by treatment type, which provided critical optimization opportunities.

Revenue Analysis:

Key insights:

- The hospital lost £37,218.95 in 2023
- Only 32.9% of bills were paid
- Failed payments on completed appointments cost the hospital £48,629.80

The first action was to find the cumulative profit of the hospital in the year 2023, which will be calculated by taking the cumulative cost away from the cumulative revenue. This began by finding the cumulative revenue by month.

To work out the cumulative revenue per treatment per month, pending payments were left out since the final outcome (payout or failed payment) remains unknown. Therefore the result will only be using confirmed revenue.

The next step was to work out the cumulative cost by month. For this, the scheduled appointments were left out, so that the result is complete data rather than speculating on whether a scheduled appointment will take place or not. Examining the difference in cost to the hospital from cancelled, no-show, and completed appointments was also important. This examination determined if the recorded cost reflected an expected reduction in the cost of no-show or cancelled appointments, since the appointment did not take place.

Average cost by appointment status

Apppointment status	Sum of Average cost
Cancelled	£2,981.27
No-show	£2,743.80
Completed	£2,697.84

The table above shows that both no-show and cancelled appointments in fact cost the hospital more than completed appointments. This cannot be the case since the appointment did not take place and so a reduced cost would be expected.

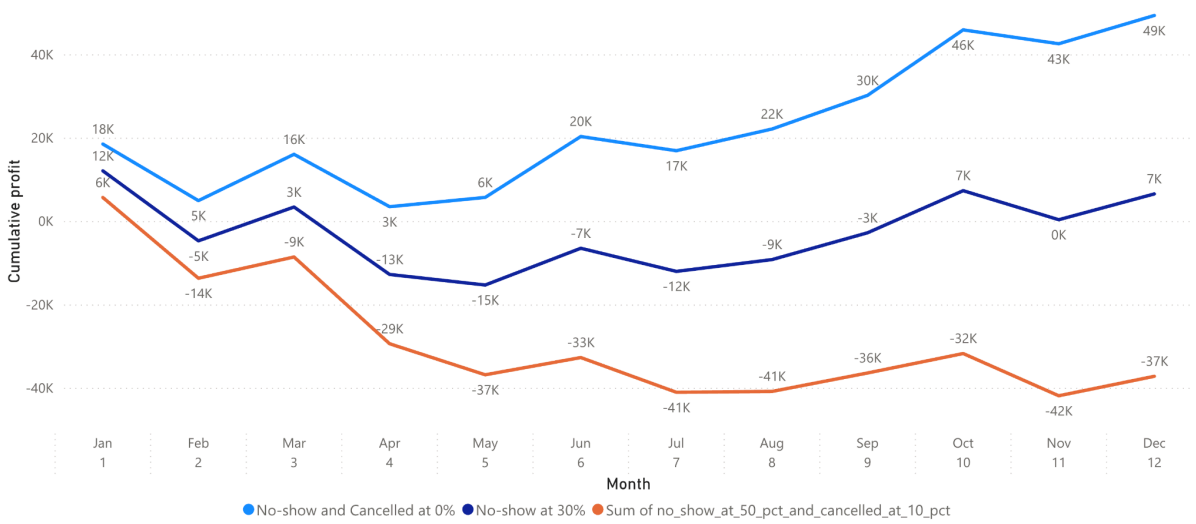
So below, an analysis shows a sample of outcomes of the 2023 profit, if the cost to the hospital of no-show and cancelled appointments vary. The analysis examined the change if both no-show and cancelled appointments cost the hospital nothing, if only no-shows costs the hospital 30% and cancelled costs nothing, and if no-show costs 50% and cancelled costs 10%.

Sum of 2023 profit with various No-show and Cancelled costs

Percentage of No-Show and Cancelled costs	Sum of 2023 Profit
No-show and Cancelled at 0%	£49,324.44
No-show at 30%	£6,521.1
No-show at 50% and Cancelled at 10%	-£37,218.95

The diagram below shows the cumulative profit by month with these changes to no-show and cancelled appointments as a line graph.

Expected cumulative profit by month with no-show and cancelled appointments at various percentages



It can be seen that if no-show and cancelled appointments do not cost the hospital anything, the profit by the end of the year would be £49,324.44. This would drop down to £6,521.10 if every 'No-show' costs the hospital only 30%. And that the hospital would have in fact lost £37,218.95 if 'No-shows' cost 50% and 'Cancelled' cost 10%.

From this point, for consistency, it was assumed that 'No-show' appointments will cost the hospital 50% of the cost, and 'Cancelled' 10%. Which means that the hospital lost £37,218.95 in 2023 and so changes are needed to bring the hospital into profit.

As a result, an analysis was conducted into how the hospital runs to find areas for optimization and revenue opportunities, to increase revenue and lower the risk with which the hospital operates under.

Price change initiatives:

Below are two possible initiatives to increase revenue. These were then simulated by applying them to the previous year, to understand the best strategies to begin taking a profit and running a successful, helpful and secure hospital.

Initiative 1: Failed payment recoupment plan

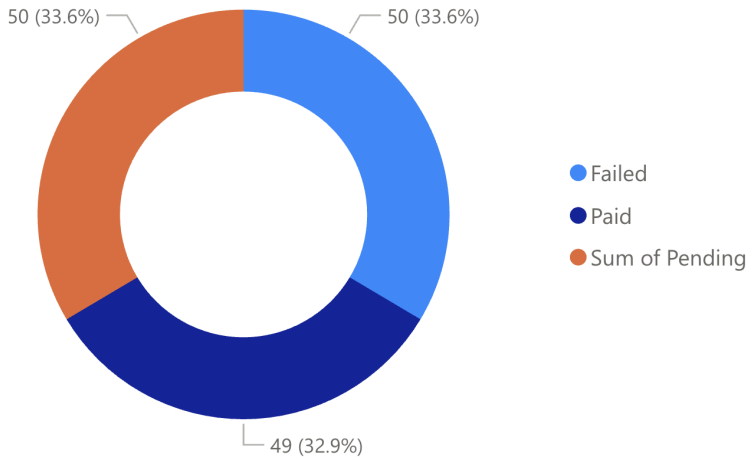
Actionable insight:

A 60% fee reduction for bills with failed payments, adopted at rate of 50% of failed bills now being paid, would lower the cumulative loss from £37,218.95 to only a loss of £5,560.23

Analysis:

The chart below represents the breakdown of paid, pending and failed payments.

Breakdown of paid, failed and pending payments amongst 'Completed', 'No-show' and 'Cancelled' appointments



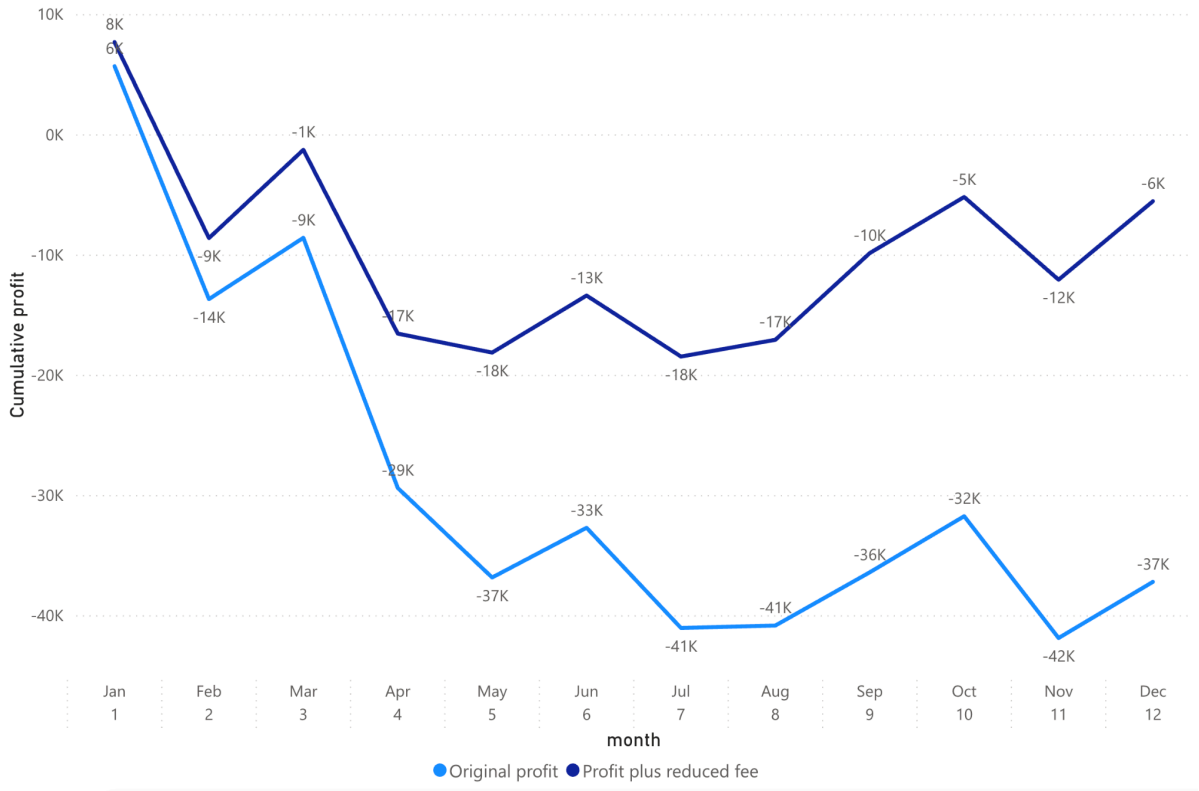
This shows that Paid payments only made up 32.9% of all bills.

Therefore, one possible initiative that could increase the growth of revenue would be to follow up on failed payments and perhaps introduce a reduced payment scheme if patients are struggling to pay their hospital bill. This could be chosen to be a temporary solution, or a more long term option with clear guidelines to go along with it.

Below, the gain in revenue if the hospital introduced a reduced fee of 40% of the total price for the patients who have failed to pay their bills was simulated .

In this simulation, it was assumed that the reduced fee will incentivise half of the failed payments to now be paid at 40%, with the other half remaining unpaid. (Scheduled appointments were excluded, and it was again assumed no-show appointments cost the hospital 50% of the cost and cancelled appointments cost 10%)

Gain in revenue with a 40% reduced fee on failed payments analysis



This illustrates how the price reduction would lower the cumulative loss from £37,218.95 to only a loss of £5,560.23.

Initiative 2: Optimized price increase

Actionable insight:

An optimised price increase of 28% would bring the yearly profit from -£37,218.95 to £11,340.02

Analysis:

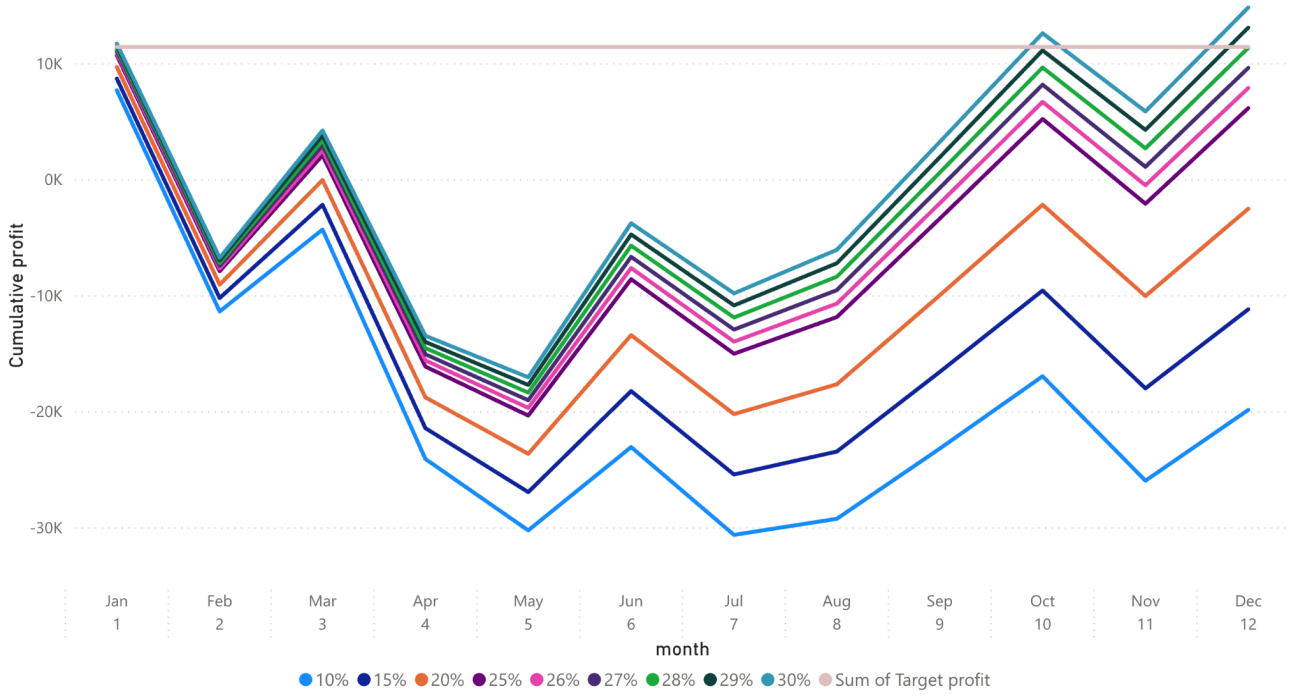
Failed payments on completed appointments cost the hospital £48,629.80 in 2023. A possible solution to mitigate the damage of these losses would be to introduce a price increase.

Cost of failed payments on completed appointments

Year	Cost of completed appointments with failed payments
2023	48,629.80

Introducing a price increase can come with risks, for example, losing patients because the cost of treatment is too high. Therefore, to find an optimal amount to raise the prices by, a few different price changes were simulated, with the aim of reducing the revenue lost from failed payments of completed appointments whilst not increasing the prices to an amount that becomes unaffordable. The aim was to increase the yearly revenue by £48,629.80, the amount lost to failed payments on completed appointments. Since the yearly profit was a loss of £37,218.95, a new yearly profit of around £11,410.85 became the target. This strategy aimed to find a price increase that reduces the risk to the hospital of the lost revenue from unpaid completed appointments, whilst keeping the increase as affordable as possible.

Cost of failed payments on completed appointments

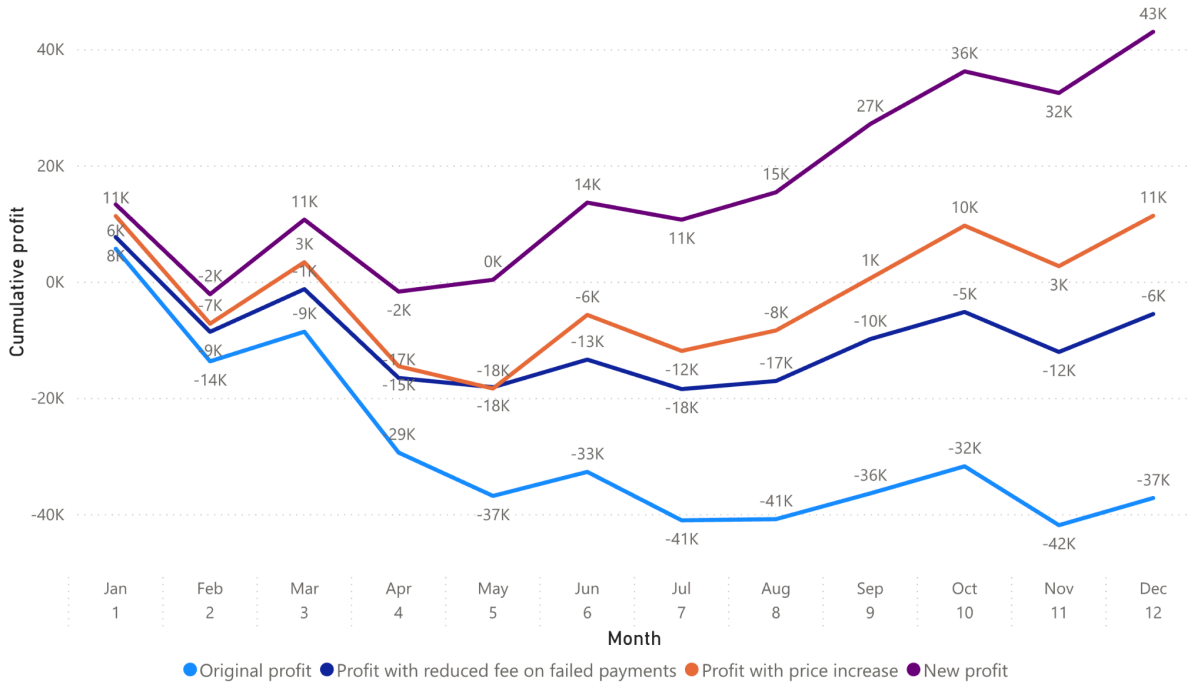


Here, the target profit line of £11,340.02 intersects with the simulated profit of a 28% price increase, and therefore this is the most optimal price increase. This price increase would bring the yearly profit from -£37,218.95 to £11,340.02.

Combining the two initiatives:

The cumulative profit in the year 2023 if both the price increase and the reduced fee on failed payments were implemented is simulated.

Cumulative profit with both initiatives versus without



This illustrates how if both initiatives are implemented then there will be an increase in revenue of £80,217.69, and the profit in the year 2023 will have gone from -£37,218.95 to £42,998.74.

Revenue by Treatment type:

Actionable insight:

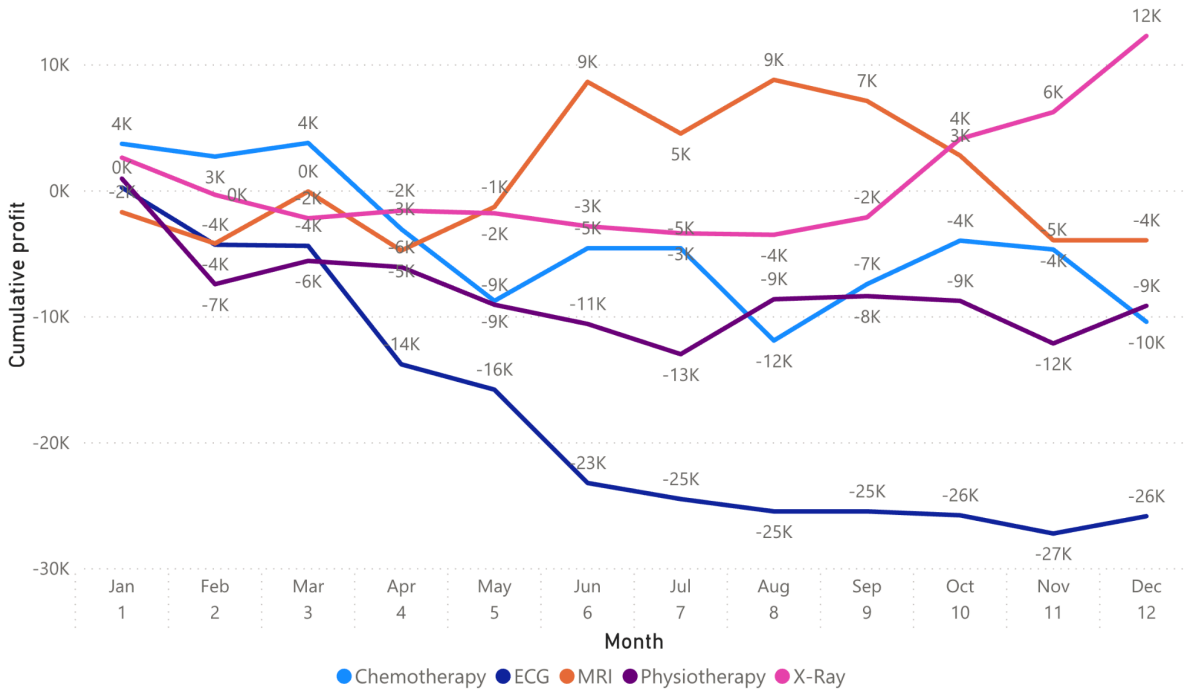
The X-Ray department needs to be further analysed to understand why it was the only department to bring in a profit. The ECG department also needs further analysis since it had the largest loss, around £16,000 less than the next biggest loss.

Analysis:

The next area of analysis was the revenue and profit of each treatment type. The target was to find which areas were succeeding and which were not, with the goal of finding where a deeper analysis of department should be made

First, the cumulative profit of each treatment type by month was analysed through a line graph.

Cumulative profit of each treatment type by month



This graph shows that the only treatment to bring in a profit in the year 2023 was X-Ray, with a yearly profit of £12,251.43. There also appeared to be an increase in the profit of X-Ray from September to the end of the year, so it would be worth analysing this further to find what caused this increase.

ECG had the worst year, losing £25,880.72. From this, the hospital needs to analyse both X-Ray and ECG much deeper, to understand why there is such a disparity between the two profits.

It also appears that MRI had a boost in profit from the months of April to August. Which again would be worth researching deeper to understand what trends led to this boost in profit.

Insurance providers pay-out rate:

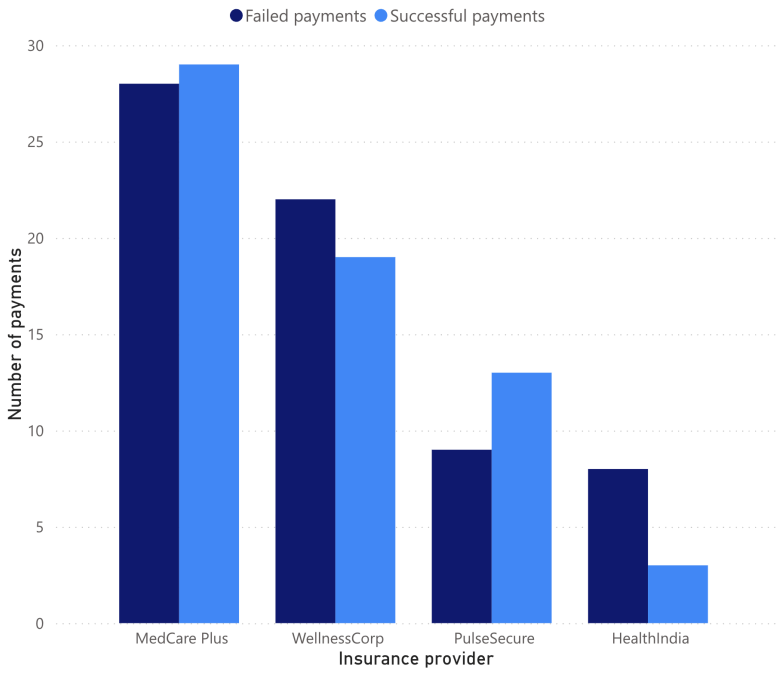
Actionable insight:

Option to offer WellnessCorp and MedCare Plus an incentivised contract, to try and increase their cover of the hospital's patients, since they had the two-highest payout rates. Also, could reduce coverage with HealthIndia since they have a very low payout rate of only 27%.

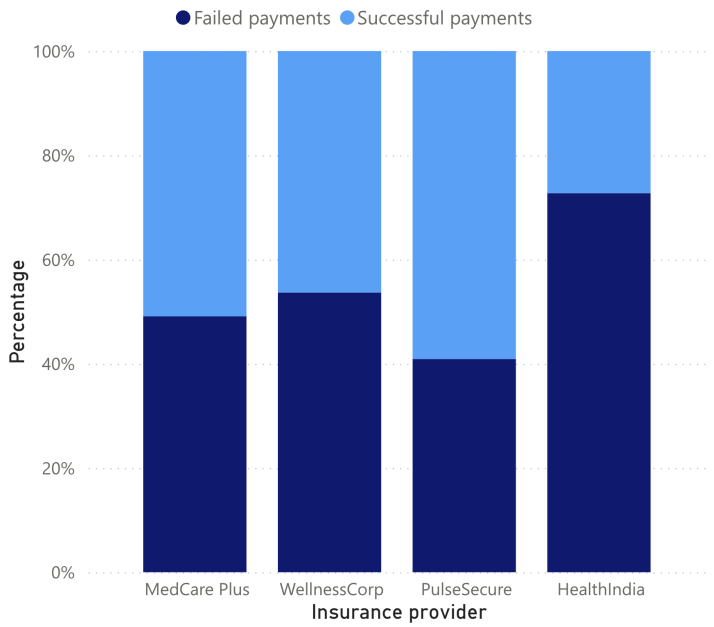
Analysis:

This next section focuses on insurance providers pay-out rate, with the aim of finding the most reliable providers.

Successful and failed payments by insurance provider



Percentage of successful and failed payments



This chart shows that 'PulseSecure' and 'MedCare Plus' are the only two providers with more successful payments than failed payments. Also, 'Medcare Plus' has the most successful payments. So, one possible suggestion from this would be to offer these two insurance providers a slightly reduced fee, with the goal of incentivising more coverage from them and increasing the overall expected pay-out rate of the hospital.

Also, 'HealthIndia' only has a successful payment percentage of 27%. One possible suggestion would be that the hospital could offer less payments with 'HealthIndia', to reduce unpaid appointments.

Analysing high-revenue earning doctors:

Actionable insight:

Remove scheduled appointments from doctors calendars if the payment has already failed, or follow up on the failed payment before the appointment takes place. This should free up time for the doctors to give more coverage to more patients within the hospital.

Analysis:

The next aim was trying to find the highest revenue earning doctors. Below is a table showing the number of completed appointments each doctor had in the year 2023 and the expected revenue per appointment. Then, each doctor was placed into a group of either 'High', 'Medium', or 'Low' to help analyse which are the highest revenue earning doctors.

Doctors and their average revenue per appointment

Doctor ID	Number of appointments	Expected revenue	Average revenue per appointment	Revenue grouping
D007	5	17,479.42	3,495.88	High
D006	5	17,189.78	3,437.96	High
D004	3	9,905.86	3,301.95	High
D008	4	12,268.14	3,067.03	High
D001	6	18,234.15	3,039.03	High
D002	5	14,641.51	2,928.30	Medium
D005	4	8,744.70	2,186.18	Medium
D003	6	11,838.40	1,973.07	Low
D010	5	9,381.52	1,876.30	Low
D009	3	4,416.98	1,472.33	Low

This shows that there are five doctors with an average revenue per appointment of over £3,000. One possible option to increase revenue would be to increase these doctors appointments. To make this possible, appointments with these doctors that were scheduled but the payments had failed could be removed from their schedules. This means that those free spaces on their calendars could be filled by new appointments with new patients.

Below is the combined table showing how many scheduled appointments can be cancelled for each doctor, as well as what revenue grouping they are in (High, Medium, or Low).

Doctors and their number of scheduled appointments that can be cancelled, as well as what revenue grouping they are in

Doctor ID	Scheduled appointments with failed payments	New appointment slots	Revenue per appointment
D005	3	High	Medium
D010	3	High	Low
D003	2	Medium	Low
D004	2	Medium	High
D006	2	Medium	High
D001	1	Low	High
D002	1	Low	Medium
D007	1	Low	High
D008	1	Low	High
D009	1	Low	Low

From this, we can see that doctors D005, D004, D006 all have a high to medium level in both; new appointment slots, and revenue per appointment, and so removing those slots to free up space in their schedules would increase revenue. Therefore, if the scheduled appointments with already failed payments are removed from all the doctors schedules, there will be an increase in revenue.

SQL script in order of appearance in report

Hospital 2023 revenue dashboard

```sql
SELECT
  FORMAT_DATE('%b', b.bill_date) AS month,
  t.treatment_type,

-- Doughnut Chart
        COUNT(CASE
           WHEN b.payment_status = 'Paid' THEN 1
           ELSE NULL END) AS paid_appointments,
        COUNT(CASE
           WHEN b.payment_status = 'Failed' THEN 1
           ELSE NULL END) AS failed_appointments,
        COUNT(CASE
           WHEN b.payment_status = 'Pending' THEN 1
           ELSE NULL END) AS pending_appointments,

-- Line graph
  ROUND(SUM(SUM(
        (CASE
           WHEN b.payment_status = 'Paid' THEN b.amount
           ELSE 0 END)
         - (CASE
           WHEN a.status = 'Completed' THEN t.cost
           WHEN a.status = 'No-show' THEN 0.5 * t.cost
           WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
            ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS cumulative_profit,

-- Cards
  ROUND(SUM
        (CASE
           WHEN b.payment_status = 'Paid' THEN b.amount
           ELSE 0 END), 2) AS revenue_by_month,
  ROUND(SUM(CASE
           WHEN a.status = 'Completed' THEN t.cost
           WHEN a.status = 'No-show' THEN 0.5 * t.cost
           WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
            ELSE 0
        END), 2) AS cost_by_month,
```

```sql
  ROUND(SUM(
        (CASE
           WHEN b.payment_status = 'Paid' THEN b.amount
           ELSE 0 END)
         - (CASE
           WHEN a.status = 'Completed' THEN t.cost
           WHEN a.status = 'No-show' THEN 0.5 * t.cost
           WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
            ELSE 0
        END)), 2) AS profit_by_month,

-- Bar chart
  COUNT(
        (CASE
           WHEN b.payment_status = 'Paid' THEN 1
           ELSE NULL END)) AS paid,
  COUNT(
        (CASE
           WHEN b.payment_status != 'Paid' THEN 1
           ELSE NULL END)) AS unpaid

FROM billing b
FULL JOIN treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN appointments a
ON t.appointment_id = a.appointment_id
GROUP BY EXTRACT(MONTH FROM b.bill_date), FORMAT_DATE('%b', b.bill_date), t.treatment_type
ORDER BY EXTRACT(MONTH FROM b.bill_date);
```

Average cost by appointment status

```sql
SELECT a.status AS appointment_status, ROUND(AVG(b.amount), 2) AS average_cost
FROM billing b
FULL JOIN treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN appointments a
ON t.appointment_id = a.appointment_id
WHERE a.status != 'Scheduled'
GROUP BY a.status
ORDER BY average_cost DESC;
```

Sum of 2023 profit with various No-show and Cancelled costs

```
SELECT
   ROUND(SUM((CASE WHEN b.payment_status = 'Paid' THEN b.amount ELSE 0 END)
       - (CASE WHEN a.status = 'Completed' THEN t.cost ELSE 0 END)), 2) AS
no_show_and_cancelled_at_0_pct,
   ROUND(SUM((CASE WHEN b.payment_status = 'Paid' THEN b.amount ELSE 0 END)
       - (CASE
          WHEN a.status = 'Completed' THEN t.cost
          WHEN a.status = 'No-show' THEN 0.3 * t.cost
          ELSE 0 END)), 2) AS no_show_at_30_pct,
   ROUND(SUM((CASE WHEN b.payment_status = 'Paid' THEN b.amount ELSE 0 END)
       - (CASE
          WHEN a.status = 'Completed' THEN t.cost
          WHEN a.status = 'No-show' THEN 0.5 * t.cost
          WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
          ELSE 0 END)), 2) AS no_show_at_50_pct_and_cancelled_at_10_pct
FROM brave-tea-484509-g6.hospital_management.billing b
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN brave-tea-484509-g6.hospital_management.appointments a
ON t.appointment_id = a.appointment_id;
```

Expected cumulative profit by month with no-show and cancelled appointments at various percentages

```
SELECT
  FORMAT_DATE('%b', b.bill_date) AS month,
  ROUND(SUM(SUM(
       (CASE
          WHEN b.payment_status = 'Paid' THEN b.amount
          ELSE 0
       END) - (CASE
          WHEN a.status = 'Completed' THEN t.cost
```

```
        ELSE 0
      END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
no_show_and_cancelled_at_0_pct,
  ROUND(SUM(SUM(
      (CASE
        WHEN b.payment_status = 'Paid' THEN b.amount
        ELSE 0
      END) - (CASE
        WHEN a.status = 'Completed' THEN t.cost
        WHEN a.status = 'No-show' THEN 0.3 * t.cost
        ELSE 0
      END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
no_show_at_30_pct,
  ROUND(SUM(SUM(
      (CASE
        WHEN b.payment_status = 'Paid' THEN b.amount
        ELSE 0
      END) - (CASE
        WHEN a.status = 'Completed' THEN t.cost
        WHEN a.status = 'No-show' THEN 0.5 * t.cost
        WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
        ELSE 0
      END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
no_show_at_50_pct_and_cancelled_at_10_pct,
FROM brave-tea-484509-g6.hospital_management.billing b
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN brave-tea-484509-g6.hospital_management.appointments a
ON t.appointment_id = a.appointment_id
GROUP BY EXTRACT(MONTH FROM b.bill_date), FORMAT_DATE('%b', b.bill_date)
ORDER BY EXTRACT(MONTH FROM b.bill_date);
```

Breakdown of paid, failed and pending payments amongst 'Completed', 'No-show' and 'Cancelled' appointments

```
SELECT
  COUNT(CASE
    WHEN a.status != 'Scheduled' AND b.payment_status = 'Paid' THEN 1
  END) AS paid,
  COUNT(CASE
    WHEN a.status != 'Scheduled' AND b.payment_status = 'Failed' THEN 1
```

```
  END) AS failed,
  COUNT(CASE
    WHEN a.status != 'Scheduled' AND b.payment_status = 'Pending' THEN 1
  END) AS pending
FROM brave-tea-484509-g6.hospital_management.billing b
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN brave-tea-484509-g6.hospital_management.appointments a
ON t.appointment_id = a.appointment_id;
```

Gain in revenue with a 40% reduced fee on failed payments analysis

```
SELECT
  FORMAT_DATE('%b', b.bill_date) AS month,
  ROUND(SUM(SUM(
      (CASE
        WHEN b.payment_status = 'Paid' THEN b.amount
        ELSE 0
      END) - (CASE
        WHEN a.status = 'Completed' THEN t.cost
        WHEN a.status = 'No-show' THEN 0.5 * t.cost
        WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
        ELSE 0
      END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS profit,
  ROUND(SUM(SUM(
      (CASE
        WHEN b.payment_status = 'Paid' THEN b.amount
        WHEN b.payment_status = 'Failed' AND a.status IN ('Completed', 'No-show', 'Cancelled')
THEN 0.5 * 0.4 * b.amount
        ELSE 0
      END) - (CASE
        WHEN a.status = 'Completed' THEN t.cost
        WHEN a.status = 'No-show' THEN 0.5 * t.cost
        WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
        ELSE 0
      END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
profit_plus_reduced_fee
FROM brave-tea-484509-g6.hospital_management.billing b
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
```

```
ON b.treatment_id = t.treatment_id
FULL JOIN brave-tea-484509-g6.hospital_management.appointments a
ON t.appointment_id = a.appointment_id
GROUP BY EXTRACT(MONTH FROM b.bill_date), FORMAT_DATE('%b', b.bill_date)
ORDER BY EXTRACT(MONTH FROM b.bill_date);
```

Cost of failed payments on completed appointments

```
SELECT FORMAT_DATE('%Y', b.bill_date) AS year,
  ROUND(SUM(CASE
    WHEN b.payment_status = 'Failed' AND a.status = 'Completed' THEN b.amount
    ELSE 0
  END), 2) AS failed_payment_on_completed_appointment
FROM brave-tea-484509-g6.hospital_management.appointments a
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON a.appointment_id = t.appointment_id
FULL JOIN brave-tea-484509-g6.hospital_management.billing b
ON t.treatment_id = b.treatment_id
GROUP BY Year;
```

Cost of failed payments on completed appointments

```
SELECT
  FORMAT_DATE('%b', b.bill_date) AS month,
  ROUND(SUM(SUM(
        (CASE
           WHEN b.payment_status = 'Paid' THEN 1.1 * b.amount
           ELSE 0
        END) - (CASE
           WHEN a.status = 'Completed' THEN t.cost
           WHEN a.status = 'No-show' THEN 0.5 * t.cost
           WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
            ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
ten_pct_price_increase,
  ROUND(SUM(SUM(
        (CASE
           WHEN b.payment_status = 'Paid' THEN 1.15 * b.amount
           ELSE 0
        END) - (CASE
           WHEN a.status = 'Completed' THEN t.cost
```

```
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
          END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
fifteen_pct_price_increase,
  ROUND(SUM(SUM(
          (CASE
            WHEN b.payment_status = 'Paid' THEN 1.2 * b.amount
            ELSE 0
          END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
          END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
twenty_pct_price_increase,
  ROUND(SUM(SUM(
          (CASE
            WHEN b.payment_status = 'Paid' THEN 1.25 * b.amount
            ELSE 0
          END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
          END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
twenty_five_pct_price_increase,
  ROUND(SUM(SUM(
          (CASE
            WHEN b.payment_status = 'Paid' THEN 1.26 * b.amount
            ELSE 0
          END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
          END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
twenty_six__pct_price_increase,
  ROUND(SUM(SUM(
          (CASE
            WHEN b.payment_status = 'Paid' THEN 1.27 * b.amount
```

```
            ELSE 0
        END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
twenty_seven__pct_price_increase,
  ROUND(SUM(SUM(
        (CASE
            WHEN b.payment_status = 'Paid' THEN 1.28 * b.amount
            ELSE 0
        END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
twenty_eight__pct_price_increase,
  ROUND(SUM(SUM(
        (CASE
            WHEN b.payment_status = 'Paid' THEN 1.29 * b.amount
            ELSE 0
        END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
twenty_nine__pct_price_increase,
  ROUND(SUM(SUM(
        (CASE
            WHEN b.payment_status = 'Paid' THEN 1.3 * b.amount
            ELSE 0
        END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
thirty__pct_price_increase
```

```
FROM brave-tea-484509-g6.hospital_management.billing b
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN brave-tea-484509-g6.hospital_management.appointments a
ON t.appointment_id = a.appointment_id
GROUP BY EXTRACT(MONTH FROM b.bill_date), FORMAT_DATE('%b', b.bill_date)
ORDER BY EXTRACT(MONTH FROM b.bill_date);
```

Cumulative profit with both initiatives versus without

```
SELECT
  FORMAT_DATE('%b', b.bill_date) AS month,
  ROUND(SUM(SUM(
      (CASE
        WHEN b.payment_status = 'Paid' THEN b.amount
        ELSE 0
      END) - (CASE
        WHEN a.status = 'Completed' THEN t.cost
        WHEN a.status = 'No-show' THEN 0.5 * t.cost
        WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
        ELSE 0
      END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS original_profit,
  ROUND(SUM(SUM(
      (CASE
        WHEN b.payment_status = 'Paid' THEN b.amount
        WHEN b.payment_status = 'Failed' AND a.status IN ('Completed', 'No-show', 'Cancelled')
THEN 0.5 * 0.4 * b.amount
        ELSE 0
      END) - (CASE
        WHEN a.status = 'Completed' THEN t.cost
        WHEN a.status = 'No-show' THEN 0.5 * t.cost
        WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
        ELSE 0
      END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
profit_with_reduced_fee_on_failed_payments,
  ROUND(SUM(SUM(
      (CASE
        WHEN b.payment_status = 'Paid' THEN 1.28 * b.amount
        ELSE 0
      END) - (CASE
        WHEN a.status = 'Completed' THEN t.cost
```

```
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
            ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
profit_with_price_increase,
  ROUND(SUM(SUM(
        (CASE
            WHEN b.payment_status = 'Paid' THEN 1.28 * b.amount
            WHEN b.payment_status = 'Failed' AND a.status IN ('Completed', 'No-show', 'Cancelled')
THEN 0.5 * 0.4 * b.amount
            ELSE 0
        END) - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
            ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS new_profit
FROM brave-tea-484509-g6.hospital_management.billing b
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN brave-tea-484509-g6.hospital_management.appointments a
ON t.appointment_id = a.appointment_id
GROUP BY EXTRACT(MONTH FROM b.bill_date), FORMAT_DATE('%b', b.bill_date)
ORDER BY EXTRACT(MONTH FROM b.bill_date);
```

Cumulative profit of each treatment type by month

```
SELECT
  FORMAT_DATE('%b', b.bill_date) AS month,
  ROUND(SUM(SUM(
        (CASE
            WHEN t.treatment_type = 'Chemotherapy' AND b.payment_status = 'Paid' THEN
b.amount
            ELSE 0 END)
         - (CASE
```

```
            WHEN t.treatment_type = 'Chemotherapy' AND a.status = 'Completed' THEN t.cost
            WHEN t.treatment_type = 'Chemotherapy' AND a.status = 'No-show' THEN 0.5 * t.cost
            WHEN t.treatment_type = 'Chemotherapy' AND a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END)))OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
chemotherapy_monthly_profit,
  ROUND(SUM(SUM(
        (CASE
            WHEN t.treatment_type = 'ECG' AND b.payment_status = 'Paid' THEN b.amount
            ELSE 0 END)
             - (CASE
            WHEN t.treatment_type = 'ECG' AND a.status = 'Completed' THEN t.cost
            WHEN t.treatment_type = 'ECG' AND a.status = 'No-show' THEN 0.5 * t.cost
            WHEN t.treatment_type = 'ECG' AND a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END)))OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
ecg_monthly_profit,
  ROUND(SUM(SUM(
        (CASE
            WHEN t.treatment_type = 'MRI' AND b.payment_status = 'Paid' THEN b.amount
            ELSE 0 END)
             - (CASE
            WHEN t.treatment_type = 'MRI' AND a.status = 'Completed' THEN t.cost
            WHEN t.treatment_type = 'MRI' AND a.status = 'No-show' THEN 0.5 * t.cost
            WHEN t.treatment_type = 'MRI' AND a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END)))OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
mri_monthly_profit,
  ROUND(SUM(SUM(
        (CASE
            WHEN t.treatment_type = 'Physiotherapy' AND b.payment_status = 'Paid' THEN b.amount
            ELSE 0 END)
             - (CASE
            WHEN t.treatment_type = 'Physiotherapy' AND a.status = 'Completed' THEN t.cost
            WHEN t.treatment_type = 'Physiotherapy' AND a.status = 'No-show' THEN 0.5 * t.cost
            WHEN t.treatment_type = 'Physiotherapy' AND a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END)))OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
physiotherapy_monthly_profit,
  ROUND(SUM(SUM(
        (CASE
```

```
            WHEN t.treatment_type = 'X-Ray' AND b.payment_status = 'Paid' THEN b.amount
            ELSE 0 END)
             - (CASE
            WHEN t.treatment_type = 'X-Ray' AND a.status = 'Completed' THEN t.cost
            WHEN t.treatment_type = 'X-Ray' AND a.status = 'No-show' THEN 0.5 * t.cost
            WHEN t.treatment_type = 'X-Ray' AND a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END)))OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS
x_ray_monthly_profit,
  ROUND(SUM(SUM(
        (CASE
            WHEN b.payment_status = 'Paid' THEN b.amount
            ELSE 0 END)
         - (CASE
            WHEN a.status = 'Completed' THEN t.cost
            WHEN a.status = 'No-show' THEN 0.5 * t.cost
            WHEN a.status = 'Cancelled' THEN 0.1 * t.cost
             ELSE 0
        END))) OVER (ORDER BY MIN(EXTRACT(MONTH FROM b.bill_date))), 2) AS cumulative_profit
FROM brave-tea-484509-g6.hospital_management.billing b
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON b.treatment_id = t.treatment_id
FULL JOIN brave-tea-484509-g6.hospital_management.appointments a
ON t.appointment_id = a.appointment_id
GROUP BY EXTRACT(MONTH FROM b.bill_date), FORMAT_DATE('%b', b.bill_date)
ORDER BY EXTRACT(MONTH FROM b.bill_date);
```

Successful and failed payments by insurance provider

And

Percentage of successful and failed payments

```
SELECT insurance_provider,
    COUNT(CASE WHEN payment_status = 'Paid' THEN 1 ELSE NULL END) AS
successful_payments,
    COUNT(CASE WHEN payment_status = 'Failed' THEN 1 ELSE NULL END) AS failed_payments,
```

```
    100 * ROUND(COUNT(CASE WHEN payment_status = 'Paid' THEN 1 ELSE NULL END) /
(COUNT(CASE WHEN payment_status = 'Failed' THEN 1 ELSE NULL END) + COUNT(CASE WHEN
payment_status = 'Paid' THEN 1 ELSE NULL END)), 2) AS successful_payment_percentage
FROM brave-tea-484509-g6.hospital_management.billing
INNER JOIN brave-tea-484509-g6.hospital_management.patients
ON billing.patient_id = patients.patient_id
GROUP BY insurance_provider
ORDER BY successful_payment_percentage DESC;
```

Doctors and their average revenue per appointment

```
SELECT a.doctor_id,
  SUM(
    CASE
      WHEN a.status = 'Completed' THEN 1
      ELSE 0
    END) appointments,
  ROUND(SUM(
    CASE
      WHEN a.status = 'Completed' THEN b.amount
    END), 2) expected_revenue,
    ROUND(ROUND(SUM(CASE WHEN a.status = 'Completed' THEN b.amount END), 2)
    / SUM(CASE WHEN a.status = 'Completed' THEN 1 ELSE 0 END), 2)
avg_revenue_per_appointment,
    CASE
      WHEN SUM(CASE WHEN a.status = 'Completed' THEN b.amount END)
    / SUM(CASE WHEN a.status = 'Completed' THEN 1 ELSE 0 END) > 3000 THEN 'High'
      WHEN SUM(CASE WHEN a.status = 'Completed' THEN b.amount END)
    / SUM(CASE WHEN a.status = 'Completed' THEN 1 ELSE 0 END) BETWEEN 2000 AND 3000
THEN 'Medium'
      ELSE 'Low'
  END AS revenue_per_appointment
FROM brave-tea-484509-g6.hospital_management.appointments a
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON a.appointment_id = t.appointment_id
FULL JOIN brave-tea-484509-g6.hospital_management.billing b
ON t.treatment_id = b.treatment_id
GROUP BY a.doctor_id
ORDER BY avg_revenue_per_appointment DESC;
```

Doctors and their number of scheduled appointments that can be cancelled, as well as what revenue grouping they are in

```
SELECT d.doctor_id,
    COUNT(CASE WHEN a.status = 'Scheduled' AND b.payment_status = 'Failed' THEN 1 END) AS
scheduled_appointments_with_failed_payments,
    CASE
      WHEN COUNT(CASE WHEN a.status = 'Scheduled' AND b.payment_status = 'Failed' THEN 1
END) >= 3 THEN 'High'
      WHEN COUNT(CASE WHEN a.status = 'Scheduled' AND b.payment_status = 'Failed' THEN 1
END) = 2 THEN 'Medium'
      WHEN COUNT(CASE WHEN a.status = 'Scheduled' AND b.payment_status = 'Failed' THEN 1
END) <=1 THEN 'Low'
    END AS freed_appointment_space,
   CASE
    WHEN SUM(CASE WHEN a.status = 'Completed' THEN b.amount END)/SUM(CASE WHEN
a.status = 'Completed' THEN 1 ELSE 0 END) > 3000 THEN 'High'
    WHEN SUM(CASE WHEN a.status = 'Completed' THEN b.amount END)/SUM(CASE WHEN
a.status = 'Completed' THEN 1 ELSE 0 END) BETWEEN 2000 AND 3000 THEN 'Medium'
    ELSE 'Low'
  END AS revenue_per_appointment
FROM brave-tea-484509-g6.hospital_management.appointments a
FULL JOIN brave-tea-484509-g6.hospital_management.doctors d
ON a.doctor_id = d.doctor_id
FULL JOIN brave-tea-484509-g6.hospital_management.treatments t
ON a.appointment_id = t.appointment_id
FULL JOIN brave-tea-484509-g6.hospital_management.billing b
ON t.treatment_id = b.treatment_id
GROUP BY doctor_id
ORDER BY (ROUND(ROUND(SUM(CASE WHEN a.status = 'Completed' THEN b.amount END),
2)/SUM(CASE WHEN a.status = 'Completed' THEN 1 ELSE 0 END), 2)) DESC;
```